Adaptive Learning, Diagnostic Assessment and Personalized Path-Planning Engine – Final Report

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Project 03 · Group 11 · CS5903 Distributed AI Training

GitHub repository	adaptive-learning-path-planner-DAT
Evidence revision	d974503 (d9745038c95717ef4e29de112227ed727045f0f2), branch `main`; final tag `cs5903-final-2026` on the commit that adds this report
Website and demo	bhuvan-56.github.io/adaptive-learning-path-planner-DAT/site/

Executive Summary

Problem. Learners differ in prior knowledge, target, deadline and daily time (5–30 min). A fixed course order cannot serve a beginner with 10 minutes a day and an experienced learner who wants a short refresher. The brief asks for an engine that estimates mastery with uncertainty, respects mandatory prerequisites, plans day by day within a time budget, adapts difficulty, schedules review and remediation, replans after missed work or changed constraints, and explains every decision, without private learner data.

What we built. A Python backend service with eight modules (M1–M8): a validated curriculum DAG; Bayesian Knowledge Tracing (BKT) mastery with an evidence-count uncertainty proxy and a four-state classifier; a 10-item computerized adaptive test (2PL IRT); a prerequisite-safe, budget-bounded daily planner with feasibility proofs and event-driven replanning; a 2-up/1-down difficulty staircase; Half-Life Regression review weighted by prerequisite importance, plus remediation; a machine-checked audit trace; and a closed-loop evaluation harness with synthetic learners. A static website with live in-browser demos runs the same Python code (Section 3.7).

Results. 21 acceptance groups: 20 pass, 1 partial (M4-a, agreed with the mentor); the recorded suite is 299 passed. Across 45,000 simulated learner-runs there are 0 prerequisite and 0 budget violations, and trace coverage is 100% for the primary arms. The adaptive arm beats the static prerequisite-order baseline in all five scenarios, by $+0.045$ to $+0.727$ concepts mastered (paired 95% CIs above zero), and improves retained recall everywhere. Greedy scheduling reaches 0.974 of the exact optimum on average (0.981 with the best-singleton fallback). The CAT is more accurate than fixed or random items in 5 of 6 paired comparisons.

Limits. All outcomes are simulator behaviour on synthetic learners. Since the diagnostic was connected to the planner (Week 8), a cold-start ablation beats the full system on mastery in all five runs (Section 5.7); remediation never triggers inside the closed loop.

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1. Problem, Scope and Success Criteria

1.1 Problem Statement

Given a curriculum of concepts with prerequisites, a learner's answers so far, a target concept, a deadline and a daily study budget, the engine must decide each day which lessons, exercises, reviews and remediation to schedule. The plan must never place a concept before its mandatory prerequisites are actually learned, must fit the day's minutes, must change when new evidence arrives or the learner misses work, and must say why every concept was included or left out.

1.2 Actors and Boundary

TABLE 1. Actors and their interaction with the engine (docs/PROBLEM_BOUNDARY.md).

Actor	**Role**	**Interface**
Learner	Answers diagnostic items, exercises and quizzes; has a target, deadline D and budget B	Scored responses (concept, correct, timestamp); receives a daily plan
Mentor / instructor	Audits why a lesson was chosen or skipped	JSON audit trace, website
Curriculum administrator	Supplies concepts, durations, difficulty, prerequisite edges and importance	Curriculum JSON, validated on load
Assessment engine (M2)	Cold-start diagnostic; flags repeated failures	CAT item selection; ability $\rightarrow$ concept priors
Planning engine (M3–M6)	Daily schedule, review, remediation, replanning, feasibility	`plan.json`, feasibility verdict, trace

1.3 Scope

In scope: the backend learner-state, diagnostic, planning, replanning, review/remediation and explanation service; transparent learner simulators (matched BKT, forgetting, mismatched PFA); a synthetic 2PL item bank and two hand-made curricula (4-concept toy, 16-concept maths); closed-loop comparative evaluation; and a static demonstration website.

Out of scope: a learning management system or learning/solving front end; curriculum content authoring; natural-language content generation; and collection of private or real student telemetry. A public-dataset fit (Junyi/ASSISTments) was deferred (Section 7.4).

1.4 Assumptions and Constraints

- Prerequisites are mandatory and the curriculum is a DAG; an activity block (a concept's lesson) is indivisible within a day.
- Durations are integer-minute proxies (ADR-006); daily budget $B \in \{5, 10, 15, 30\}$ minutes; deadline $D \geq 1$ day.
- BKT parameters are declared defaults (slip 0.10, guess 0.25, learn 0.10), not fitted; the HLR half-life is a fixed 14-day prior.
- Planned, completed and evidence-Mastered are different states: only evidence-Mastered prerequisites unlock dependents (Section 3.3).

1.5 Success Criteria

Milestones M1–M8 from the brief were refined into 21 frozen acceptance groups (M1-a … M8-d, docs/ACCEPTANCE_GROUPS.md; TC-IDs in TRACEABILITY.md; protocol in docs/EVALUAT

ION_PROTOCOL.md). Four are hard invariants that must hold in every run: **zero prerequisite violations** (M1-d), **zero budget violations** (M3-a), **trace coverage** $= 1.0$ (M7-a) and **one runnable end-to-end command** (M8-b).

The brief's "done when" is tested as: (1) learner profiles with different skills and budgets receive different valid plans; (2) plans change after new evidence or a missed day without losing completed work; (3) the adaptive system improves on static and greedy baselines in concepts mastered and retained recall, where a claim requires the **paired 95% bootstrap CI (**$B = 1{,}000$**) to exclude zero**, otherwise we report "no difference shown"; (4) every inclusion, omission, review and difficulty change carries a reason code. The scheduling target is a mean greedy/oracle ratio ≥ 0.95 on 250 instances; no worst-case guarantee is claimed.

2. Related Work, Baselines and Alternatives

2.1 Relation to Existing Work

TABLE 2. Prior work and how the final system uses it (Mid-Review Section B.4, updated).

Area	**Reference**	**How it is used at the end of the project**
Knowledge tracing	Corbett and Anderson (1994)	BKT is the mastery model for every concept (M1).
BKT implementation	Badrinath, Wang and Pardos (2021), pyBKT 1.4.3	Reference for parity tests: our posterior matches pyBKT within 10^{-4} (M1-b). Empirical fitting not done.
Deep knowledge tracing	Piech et al. (2015)	Rejected as primary model: its latent state cannot give auditable prerequisite thresholds (ADR-001).
Adaptive testing	van der Linden (2005); Eggen (2018); catsim	Implemented: 2PL CAT with maximum-information selection, 10-item cap, 24-item synthetic bank (M2, ADR-002).
Learning-path recommendation	Chen et al. (AAAI 2023)	Motivates concept-aware paths; the neural generator is replaced by an explicit DAG scheduler with hard guarantees (ADR-003).
Spaced repetition	Settles and Meeder (2016); Duolingo HLR	Implemented recall decay $R = 2^{-\Delta t/H}$ with a declared prior $H = 14$ days; review when $R < 0.85$ (M5, ADR-004).
Learner data	Chang, Hsu and Chen (2015), Junyi; Feng, Heffernan and Koedinger (2009), ASSISTments	Deferred: logs lack latent truth, prerequisite DAGs and durations; simulator-first route chosen (Section 7.4).

2.2 Alternatives Considered and Decisions

TABLE 3. Design decisions (decisions, ADR-001 to ADR-009).

ADR	Question	Options	Decision and reason	Status at end
001	Mastery model	BKT vs DKT	BKT: interpretable p, explicit evidence count, traceable thresholds	Implemented
002	Diagnostic	CAT vs fixed vs random	CAT, with fixed and random kept as measured baselines	Implemented; compared in Section 5.2
003	Planner	DAG scheduling vs set-to-sequence	DAG precedence-constrained knapsack: provable prerequisite safety	Implemented
004	Forgetting	HLR vs fixed intervals	HLR with declared prior and fallback	Implemented (H not fitted)
005	Uncertainty	Beta proxy vs full posterior	$\sigma^2 = p(1-p)/(\kappa+1)$ as evidence-count proxy, a declared heuristic	Implemented; calibration measured
006	Durations	Measured vs proxy	Integer-minute proxies per concept	Implemented
007	Conflicting evidence	Single answer vs window gate	4-answer window, 3-of-4 quorum before Mastered or Weak	Implemented
008	Self-learning policy	Bandit vs deterministic	Deterministic; bandit only inside an admissibility layer	Not implemented (no-go, W7)
009	Cold start	Ability $\rightarrow$ concept mapping	$p = \text{sigmoid}(\hat{\theta})$ for probed concepts, Beta(1,1) otherwise; W8 amendment $n = k$ answered items	Implemented

2.3 Reused Open-Source Tools

Pinned in requirements.txt: NetworkX 3.1 (BSD-3; graph storage, cycle check, topological order), pyBKT 1.4.3 (MIT; parity reference), catsim 0.21.0 (BSD-3; IRT and CAT components), NumPy 1.24.3, pandas 2.1.4, SciPy 1.11.4, scikit-learn 1.3.2 (BSD-3), PyYAML 6.0.1 (MIT), pytest 7.3.1 (MIT) and matplotlib 3.8.4. The website loads Pyodide 0.26.4 from jsDelivr (MPL-2.0). Everything else is the team's own code: the learner-state classifier and evidence gate, Q-matrix mapping, feasibility proofs, planners, replanning, trace schema and checker, simulators, retention quiz, evaluation harness and website.

2.4 Baselines

Both baselines run on the same synthetic learners, seeds and diagnostic answers as the adaptive arm, so differences are paired by learner.

- **static_prerequisite_order** (src/planning/static_baseline.py): take the target and its ancestors in lexicographic topological order and pack them day by day into the budget; never replans. This is the "static course order" the brief argues against.
- **non_adaptive_greedy** (src/planning/greedy.py): each day, among concepts whose prerequisites are done, pick by score per minute until the budget is full; no learner-state updates, no review, no replanning.

3. Architecture, Design and Implementation

3.1 Architecture

Figure 1 shows the final system. Compared with the Mid-Review architecture, every module is now implemented and the evaluation simulator sits behind an explicit latent-state boundary.

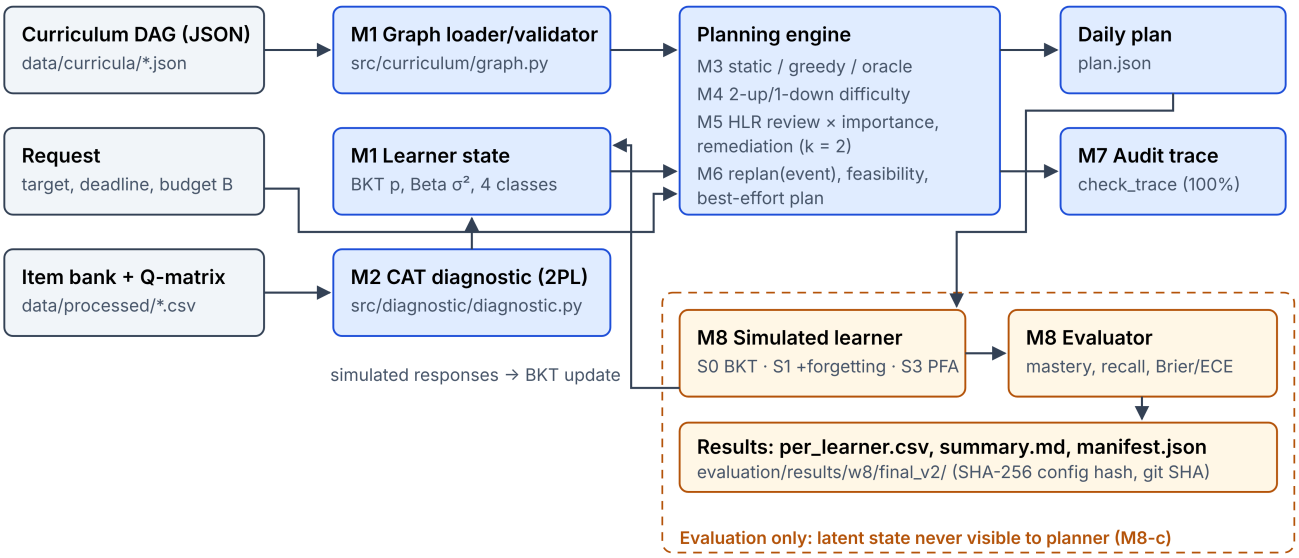


FIGURE 1. Components and data flow. Blue: engine modules. Amber: evaluation-only simulator and evaluator; their latent state is never visible to the planner (M8-c).

3.2 Major Components and Final Status

TABLE 4. Modules, interfaces and status at Week 8 (compare with Mid-Review Section C.2, status at Week 3).

Mod.	Component	Key functions / files	Status at Week 8
M1	Graph loader and validator	`load_graph`, `validate_graph` · src/curriculum/graph.py	Six checks incl. non-finite node and edge values (Mid-Review gap closed, W4 D5)
M1	Learner-state model	`ConceptState.update`, `classify`, `PolicyConfig` · src/state	BKT parity with pyBKT $< 10^{-4}$; four classes; policy schema validated
M2	Adaptive diagnostic	`run_adaptive_diagnostic`, `init_from_diagnostic` · src/diagnostic/diagnostic.py	Live CAT implemented (W4); answers count as evidence (W8, B10)
M3	Planners	`generate_static_plan`, `run_greedy_scheduler`, `run_greedy_best_singleton`, `run_exhaustive_oracle` · src/planning	Order-invariant oracle (W3); best-singleton fallback (W7)
M3/M6	Feasibility and replanning	`check_feasibility`, `replan(state, event)` · src/planning/replan.py	Four events; practice-for-evidence (W7); best-effort plan (W8)
M4	Difficulty policy	`step_difficulty` · src/planning/difficulty.py	2-up/1-down implemented (W4)

Mod.	Component	Key functions / files	Status at Week 8
M5	Review and remediation	`rank_due_reviews`, `get_remediation_ancestors` · src/planning/review.py	HLR review (W4), importance-weighted (W8, B11); remediation unit-tested
M7	Audit trace	`TraceEntry`, `ReasonCode`, `check_trace` · src/trace	11 reason codes; strict type and bound checks (W4 D4)
M8	Simulation and evaluation	`MatchedBKTGenerator`, `MismatchedPFAOrLogisticGenerator`, `run_integrated_evaluation` · src/evaluation	9-arm harness, retention quiz, calibration, paired bootstrap

3.3 Learner-State Model (M1)

BKT update. For a concept with mastery p, slip $s = 0.10$, guess $g = 0.25$ and learn rate $t = 0.10$,

$$p_{\text{post}} = \begin{cases} \dfrac{p(1-s)}{p(1-s)+(1-p)g} & \text{if the answer is correct,} \\ \dfrac{p\,s}{p\,s+(1-p)(1-g)} & \text{if it is incorrect,} \end{cases} \qquad p \leftarrow p_{\text{post}} + (1-p_{\text{post}})\,t.$$

Each answer also increments the evidence count n and the Beta concentration κ, and is appended to a window of the last $W = 4$ answers.

Uncertainty. $\sigma^2 = p(1-p)/(\kappa+1)$. This counts evidence; it is a declared heuristic, not a calibrated credible interval (ADR-005). Calibration of p is measured instead (Section 5.5).

TABLE 5. Four-state classifier, evaluated in this precedence order (src/state/learner_state.py).

State	Rule	Planner consequence
Uncertain	$n < 3$ or $\sigma^2 > 0.05$	Never trusted: routed to assessment or practice-for-evidence (`UNCERTAIN_NEEDS_ASSESSMENT`)
Mastered	$p \geq 0.85$, $\sigma^2 \leq 0.05$, $n \geq 3$ and at least 3 of the last 4 answers correct	Unlocks dependents (`ALREADY_MASTERED`); eligible for review when recall decays
Weak	$p < 0.40$ with at least 3 of the last 4 answers incorrect	Blocks dependents; candidate for remediation
Developing	otherwise	Practised; dependents wait

Planned, completed and evidence-Mastered are three different states. A concept placed on a future day is *planned*; a dependent may be placed after it only *conditionally* and is re-checked before it is done. *Completed* means the learner did the activity. Only *evidence-Mastered* unlocks dependents unconditionally; a failed prerequisite keeps its dependents blocked.

Diagnostic priors (M2 → M1). The CAT estimates ability $\hat{\theta}$ from up to 10 items chosen by maximum Fisher information from a 24-item 2PL bank, stopping early if $\text{Var}(\hat{\theta}) \leq 0.15$. Concepts probed by the Q-matrix start with $p = \text{sigmoid}(\hat{\theta})$, and since Week 8 with $n = k$ answered items, $\kappa = 2 + k$ and those answers as history (ADR-009 amendment, 70183b7); unprobed concepts start cold (Beta(1,1), Uncertain).

3.4 Planning and Adaptation Policies (M3–M6)

- **Frontier.** A concept is a candidate on day d if it is the target or an ancestor, is not evidence-Mastered or already scheduled, and its prerequisites are Mastered or planned earlier (conditional). Every candidate gets exactly one trace entry per day.
- **Greedy selection.** Candidates are ranked by score per minute with deterministic tie-breaking and packed while they fit the remaining budget. The **best-singleton fallback** (W7) also evaluates the best single item that fits and keeps whichever scores higher, the classic fix for knapsack counterexamples. The **exact oracle** enumerates precedence-closed subsets ($N \leq 20$, with a timeout) and is used to measure greedy quality.
- **Feasibility.** `INFEASIBLE_PROVED` is returned only for proved cases: a block longer than B (fix: raise the budget to the longest block; more days cannot help) or required work $W_{\text{req}} > D \times B$ (fixes: the minimum budget or number of days). `FEASIBLE` requires a witness plan; if packing fails without a proof the status is `HEURISTIC_NO_PLAN`. Since Week 8 a proved-infeasible request also returns a **best-effort plan** of the work that fits instead of idling.
- **Difficulty (M4).** Difficulty $d \in [0.05, 0.95]$: one incorrect answer lowers d by 0.10; two consecutive correct answers raise it by 0.10; an Uncertain concept returns to its curriculum baseline.
- **Spaced review (M5).** Predicted recall $R = 2^{-\Delta t/14}$. A Mastered concept is due when $R < 0.85$. Due reviews use only leftover minutes and are packed by priority $(1 - R) \times w$, where w is the largest edge importance from the concept to a not-yet-Mastered dependent (default 1.0) (`edde468`). Recall, weight and priority are written to the trace (`REVIEW_DUE`).
- **Remediation (M5).** Two consecutive failed assessments on a concept insert its lowest-mastery unmet prerequisite (`REMEDIATION_TRIGGERED`); forward progress on that branch pauses until the prerequisite is evidence-Mastered.
- **Replanning (M6).** `replan(state, event)` handles `assessment_result`, `missed_day`, `changed_budget` and `changed_deadline`. Completed work is kept and only the future is recomputed. A completed prerequisite that is still Uncertain or Weak and blocks remaining work receives one practice block (`PRACTICE_FOR_EVIDENCE`, Week 7 deadlock fix, `c5d1a8f`).

3.5 Interactions and Workflow

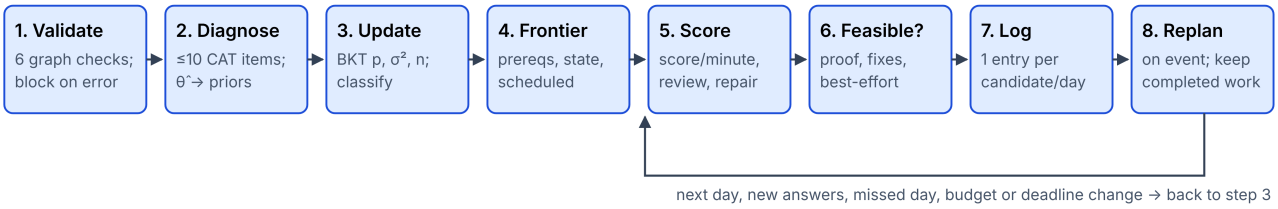


FIGURE 2. Daily workflow. Steps 3–8 repeat every day and after every event.

1. **Validate.** Any graph error blocks planning and returns a structured error instead of a fallback plan.
2. **Diagnose.** A new learner takes the bounded CAT; $\hat{\theta}$ sets the priors of probed concepts.
3. **Update.** Every scored answer updates p, σ^2, n and the class.
4. **Build the frontier** from prerequisites, states and earlier days.
5. **Score and select.** Candidates are scored; review and remediation compete for the same day's minutes.
6. **Check feasibility.** A witness plan, a proof with fixes, or a best-effort plan is returned.
7. **Log.** The trace records one disposition per candidate per day with the state and constraints at decision time.

8. **Observe and replan** on an assessment result, a missed day, or a changed budget or deadline.

3.6 Inputs, Outputs and the Audit Trace

Inputs: Curriculum JSON: nodes (`node_id`, `concept_name`, `baseline_duration`, `exercises`, `base_difficulty`) and edges (`source`, `target`, `importance`); learner responses (concept, correct, timestamp); constraints (B, D, target); and one policy configuration per run (BKT parameters, window, thresholds, seeds), hashed with SHA-256.

Outputs: `plan.json` (per day: activities, conditional activities, minutes used and left; the feasibility explanation; assessment candidates), concept states, the next difficulty, a feasibility verdict and the trace. Each trace entry has eight fields: `day`, `node_id`, `decision`, `reason_code`, `concept_state` (p, σ^2, n, κ, class), `constraints` (budget, remaining budget, prerequisites, conditional), `score_terms` and `policy_version`. A sample is in Appendix A.

TABLE 6. Reason codes (src/trace/schema.py). The Mid-Review had eight; three were added for review, remediation and practice-for-evidence.

Code	Meaning	Code	Meaning
`SELECTED`	Scheduled today	`BLOCK_EXCEEDS_DAILY_BUDGET`	Activity longer than B (proved infeasible)
`PREREQUISITE_UNMET`	A prerequisite is not Mastered or planned	`BUDGET_EXCEEDED`	Does not fit today's remaining minutes
`ALREADY_MASTERED`	Evidence-Mastered; no new learning needed	`UNCERTAIN_NEEDS_ASSESSMENT`	Not enough evidence; route to assessment
`ALREADY_SCHEDULED`	Planned on an earlier day	`REVIEW_DUE`	Mastered but predicted recall < 0.85
`COMPLETED`	Already done by the learner	`REMEDIATION_TRIGGERED`	Prerequisite inserted after two failures
`PRACTICE_FOR_EVIDENCE`	Completed but unproven prerequisite gets practice (W7)		

3.7 User Interface: Project Website and Live Demos

Purpose. The brief asks the system to "expose why a lesson was chosen". The website lets a mentor or reviewer see the engine plan, replan and explain itself without installing anything, and browse every reported result. It is a demonstration interface, not a learner-facing product (an LMS is out of scope).

How it works. A static site in site, published with GitHub Pages at bhuvan-56.github.io/adaptive-learning-path-planner-DAT/site/. The live demos run the project's own Python modules from `src/` in the browser through Pyodide 0.26.4; site/py/demo_api.py only converts JSON in and out, so nothing is re-implemented in JavaScript. If Pyodide cannot be downloaded (about 10 MB, once), the page says so and shows the saved outputs of every preset from site/data/fallback.json, which site/build_data.py produces by running the same code natively. The results pages read JSON that the same script recomputes from the per-learner CSVs; it refuses to write unless every number equals the committed result files.

TABLE 7. Website pages and what each demonstrates.

Page	**What the user can do**	**Milestones**
Overview	Engine summary (Diagnostic → Learner state → Daily plan → Replan), headline numbers with sources, Week 7–8 changes	All
Curriculum	Switch between the 16-concept maths and 4-concept toy curricula; inspect durations, difficulty, exercises and prerequisite importance; click a concept to highlight its prerequisites	M1
Live demo A: greedy vs oracle	Edit items (minutes, score, prerequisites) and a budget; run greedy, greedy + best-singleton and the exact oracle; presets reproduce the 12/18 counterexample and benchmark instance 215	M3
Live demo B: plan builder	Choose curriculum, target, budget and deadline; type each concept's answers (e.g. 1111, 0000, 1010); see learner states, the day-by-day plan drawn on the graph and the full trace with reason codes; apply one replan event	M1, M3–M7
Scheduling benchmark	Ratio distribution over 250 instances, paired difference, changed instances, saved bad cases	M3, M8
Results	S0/S1 arm tables with CIs, paired differences, the Week 8 before/after, the 2 × 2 robustness table	M8

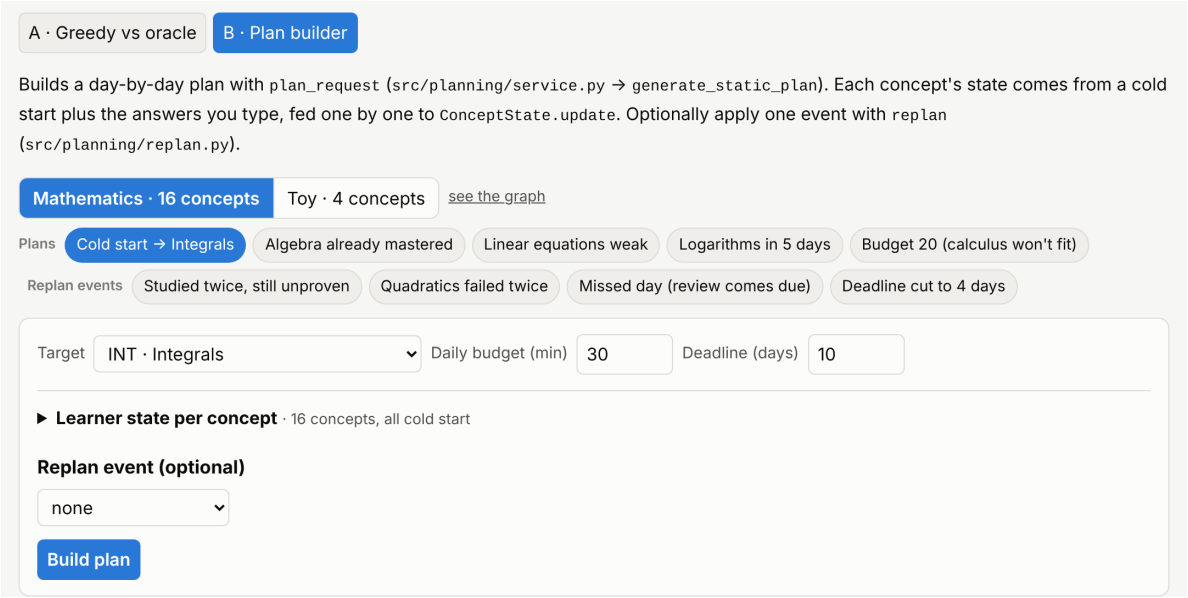


FIGURE 3. Demo B inputs: curriculum switch, plan presets ("Cold start → Integrals", "Algebra already mastered", "Budget 20 (calculus won't fit)") and replan events, target, budget and deadline, per-concept answers and an optional event.

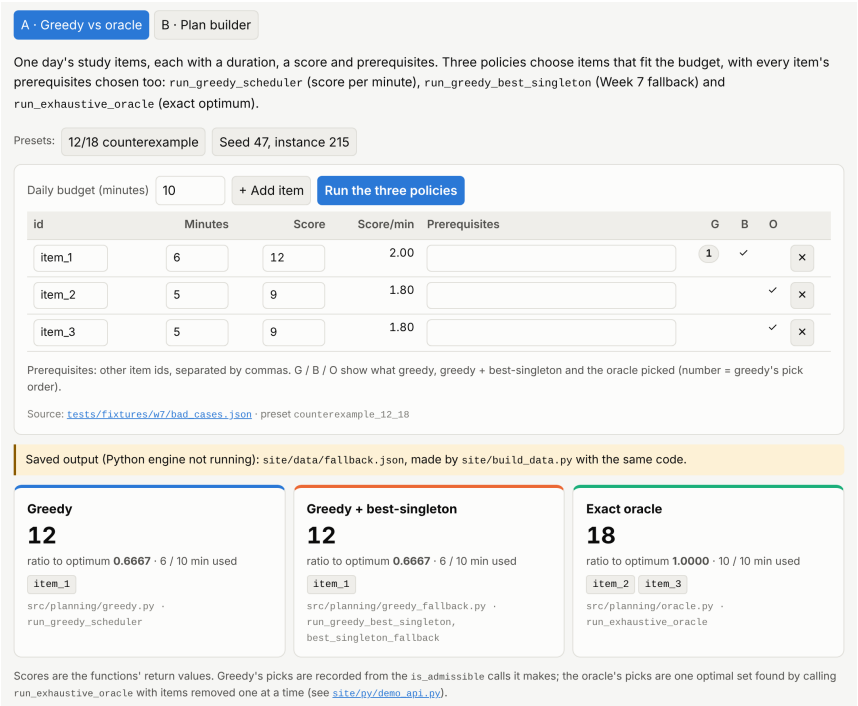


FIGURE 4. Demo A, preset "12/18 counterexample": greedy and greedy + best-singleton both score 12 (ratio 0.667); the exact oracle finds 18.

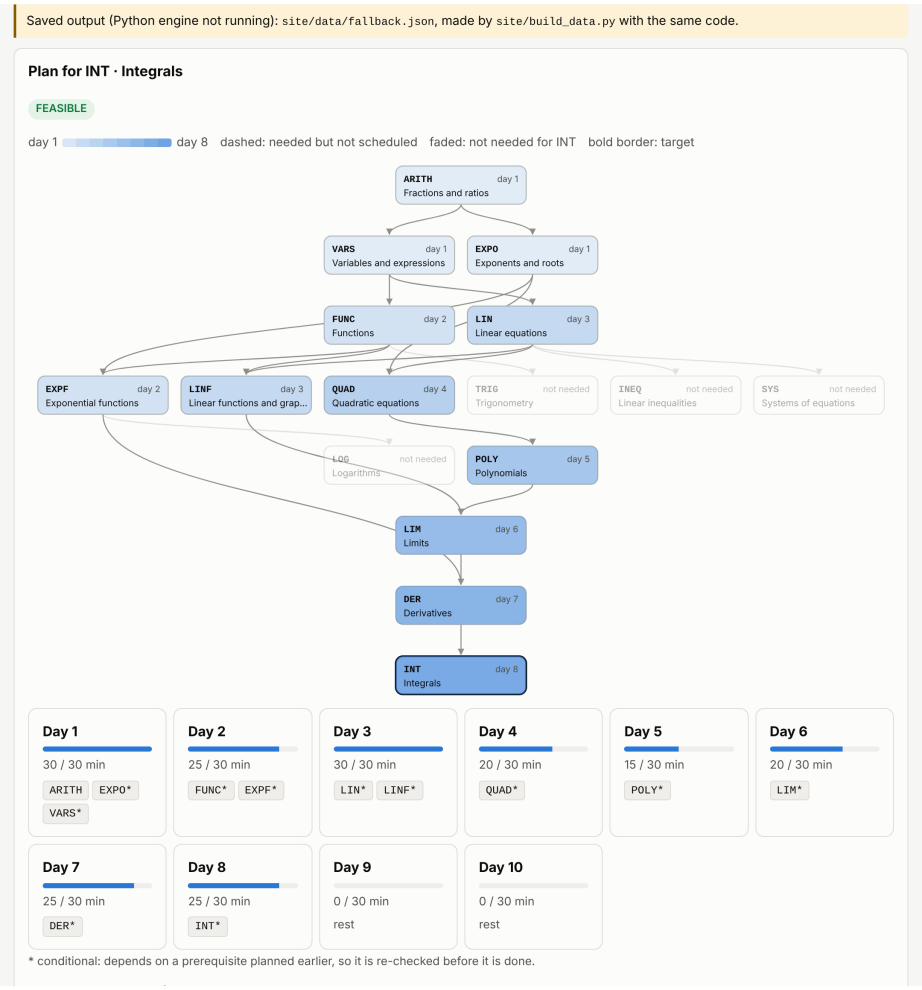


FIGURE 5. Demo B on the maths curriculum, cold start, target Integrals, 30 min/day for 10 days: status FEASIBLE; the planned day of each concept is drawn on the prerequisite graph (concepts not needed for Integrals are faded) and summarised per day below (minutes used / budget).

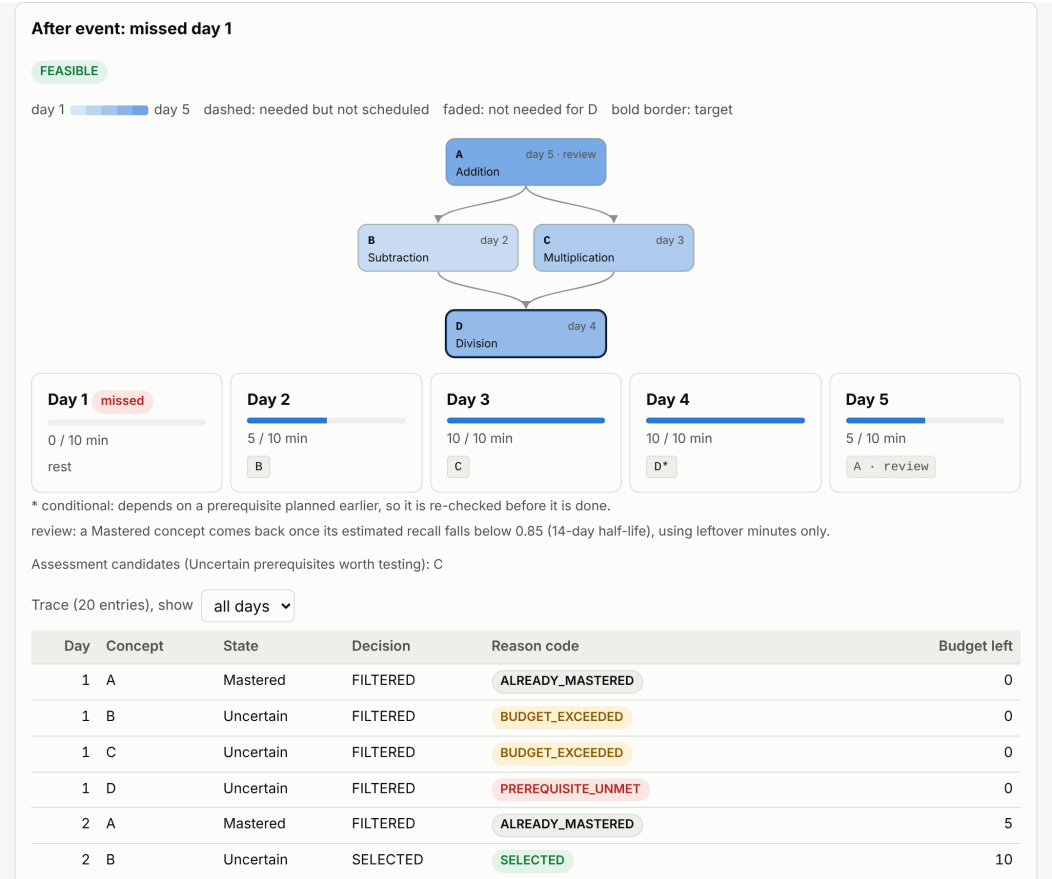


FIGURE 6. Demo B replan, toy curriculum, event "missed day 1": completed work is kept, the plan shifts by a day, and a review of Addition appears on day 5 in leftover minutes. The trace table lists every candidate with its reason code and remaining budget.

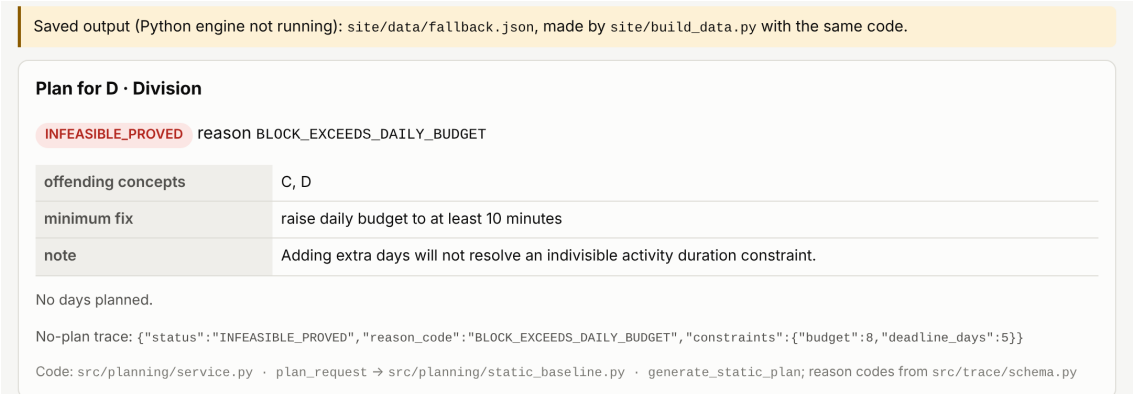


FIGURE 7. Demo B, "Budget 8 (infeasible)": `INFEASIBLE_PROVED` with reason `BLOCK_EXCEEDS_DAILY_BUDGET`, the offending concepts, the minimum fix (raise the budget to 10) and the note that extra days cannot help.

Mapping to milestones. Demo B shows M1 (states computed from typed answers), M3 (a time-budgeted, prerequisite-valid plan), M6 (replanning and infeasibility), M5 (reviews in leftover minutes) and M7 (the trace behind every decision). Demo A shows the M3 limitation that motivated the Week 7 fallback. The Results and Scheduling pages are the M8 evidence. More screenshots are in Appendix B.

Limitations. No accounts or saved learners; one event per replan; the live demos need internet access once; results pages show precomputed runs. The screenshots were captured from a local copy of the site at d974503 where the Pyodide CDN was unreachable, so the demos show the saved preset outputs (marked "Saved output" on the page); these are produced by the same code and match what the live engine returns for those presets.

3.8 Reproducibility and Provenance

Every run writes a manifest with the git SHA and dirty flag, timestamp, Python version, a SHA-256 hash of the resolved configuration, input paths and output checksums (src/provenance.py). Evaluation outputs refuse to write under `evaluation/results/` from a dirty tree. scripts/clean_run.ps1 clones the repository fresh, installs pinned dependencies, runs the suite and the demo, slice and evaluate commands, and checks every artifact (M8-b). A GitHub Actions workflow runs pytest on Ubuntu with Python 3.11.

4. Methodology and Experiment Setup

4.1 Evaluation Questions

Q1 Is the greedy planner close to optimal, and how fast is it?
Q2 Does the CAT estimate ability better than fixed or random items?
Q3 Does the adaptive system teach and retain more than static and greedy plans on the same learners?
Q4 Does the result survive forgetting, a larger curriculum and a simulator that does not match the planner's model?
Q5 Are the engine's mastery estimates calibrated?
Q6 Do the hard constraints hold everywhere?

4.2 Simulated Learners

TABLE 8. Simulator scenarios (src/simulation/simulator.py). The planner sees only answers; latent state is read by the evaluator alone.

Scenario	Learner model	Bias
S0 matched	BKT generator with the same slip, guess and learn parameters as the engine	Most favourable to the planner: learners behave exactly as it assumes
S1 forgetting	S0 plus daily forgetting $p_F = 0.05$ during study and over a 7-day delay	Planner does not model forgetting directly (B9)
S3 mismatched	PFA/logistic: accuracy $= \sigma(\beta_c + \gamma \cdot$ successes $+ \rho \cdot$ failures $-$ difficulty term); learning from practice counts	Rewards practice attempts, which favours the adaptive arm

4.3 Arms

TABLE 9. Evaluation arms (src/evaluation/integrated.py).

Arm	Priors	Planner	Difficulty	Replan	Review	Remed.
adaptive_full	CAT ($n = k$)	replan	2-up/1-down	on	on	on
static_prerequisite_order	CAT ($n = 0$)	static topological	base	off	off	off
non_adaptive_greedy	CAT ($n = 0$)	greedy score/min	base	off	off	off
non_adaptive_greedy_best_singleton	CAT ($n = 0$)	greedy + fallback	base	off	off	off

Arm	Priors	Planner	Difficulty	Replan	Review	Remed.
ablation_no_ diagnostic	cold start	replan	2-up/1-down	on	on	on
ablation_no_replan	CAT	static	2-up/1-down	off	off	off
ablation_no_ difficulty_step	CAT	replan	base	on	on	on
ablation_no_review	CAT	replan	2-up/1-down	on	off	on
ablation_no_ remediation	CAT	replan	2-up/1-down	on	on	on*

* Known defect B1: the remediation ablation does not actually disable remediation; remediation also never triggers in the loop (B2).

4.4 Workloads, Repeats and Metrics

- **Toy curriculum:** 4 concepts, target D (Division), 5 days × 10 min. **Maths curriculum:** 16 concepts, target Integrals, 10 days × 30 min.
- **Repeats:** 5 seeds (42, 101, 2024, 777, 999) × 200 learners = 1,000 paired learners per arm per run. Scheduling: 250 instances, seeds 7, 17, 27, 37, 47 (50 each).
- **Metrics:** concepts mastered (latent, at the horizon); retained recall (share correct on $N = 5$ fresh items per studied concept after a 7-day delay); no-plan days; prerequisite and budget violations, recounted independently of the planner; trace coverage; runtime p50/p95 per plan; diagnostic $|\hat{\theta} - \theta^*|$, concept Brier score and held-out log loss; engine mastery Brier score and 10-bin ECE; greedy/oracle score ratio.
- **Statistics:** per-learner paired differences; percentile bootstrap, $B = 1{,}000$; a claim only if the 95% CI excludes zero.
- **Environment:** Windows laptop, Python 3.11 virtual environment, pinned requirements; runtimes come from one machine and one run.

4.5 How to Reproduce

```
pip install -r requirements.txt
python -m pytest -q                                                  # 299 passed
python -m src.planning.benchmark                                     # Q1
python -m src.evaluation.diagnostic_comparison \
       --config experiments/w4_comparison.yaml                       # Q2
python -m src.evaluation.integrated --config experiments/w8_final.yaml    --out out/s0
python -m src.evaluation.integrated --config experiments/w8_final_s1.yaml --out out/s1
python -m src.evaluation.integrated --config experiments/w8_maths_s0.yaml --out out/m0
#   (also experiments/w8_maths_s3.yaml and experiments/w8_toy_s3.yaml)
python scripts/w7_pooled_s0.py out/s0 adaptive_full:static_prerequisite_order
powershell scripts/clean_run.ps1                                     # clean-clone audit
```

5. Results, Observations and Analysis

Raw outputs for every number below are in evaluation/results/w8/final_v2 (per-seed CSVs, summaries, manifests, `pooled_*.txt`, `before_after.txt`) unless another file is named. "*n.d.*" means no difference shown.

5.1 Scheduling Quality and Runtime (Q1)

TABLE 10. Greedy vs exact oracle on 250 synthetic instances (evaluation/results/w7/improvement/scheduling/summary.md).

Policy	Mean ratio	5th pct	Min	Below 0.95	Runtime p50 / p95
Greedy score-per-minute	0.9737	0.8270	0.2143	35 / 250	0.022 / 0.066 ms
Greedy + best-singleton	0.9814	0.8548	0.5000	25 / 250	0.027 / 0.073 ms

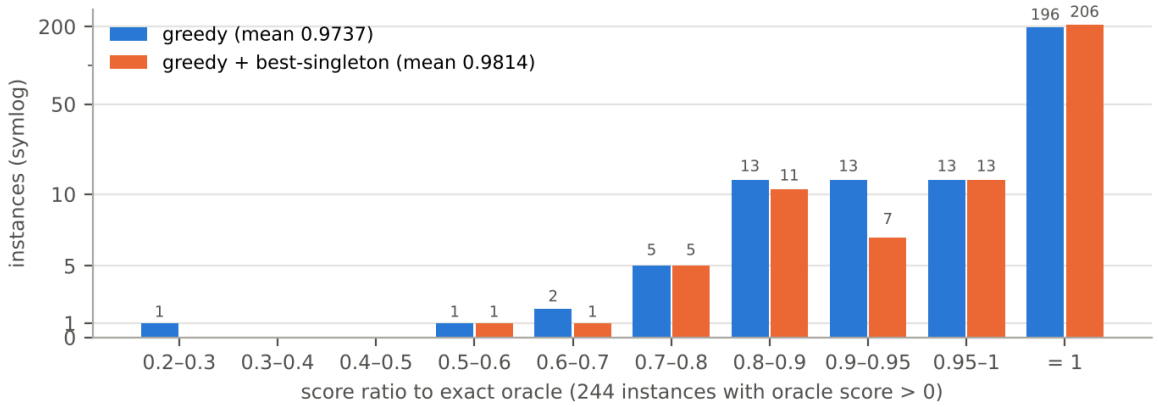


FIGURE 8. Distribution of greedy/oracle score ratios (244 instances with oracle score > 0). The fallback moves 10 instances to the optimum and lowers none.

The paired difference is $+0.0077$ [0.0018, 0.0159]: the fallback is better. The mean exceeds the 0.95 target, but individual plans can be far from optimal: instance 215 scores 0.214 with greedy and the hand-made 12/18 case stays at 0.667 with both policies, so no worst-case guarantee is claimed (B8). Mean oracle runtime is 0.24 ms on these small instances. Inside the closed loop (seed-42 summaries in `final_v2`) one adaptive plan takes 13.7 / 24.1 ms (p50 / p95) on the toy curriculum and 78.3 / 134.1 ms on the maths curriculum, against 0.56 / 0.83 ms and 2.8 / 3.8 ms for the static planner.

5.2 Diagnostic Accuracy (Q2)

TABLE 11. CAT vs fixed-order vs random items, 1,000 paired learners, all capped at 10 items (evaluation/results/w8/diagnostic_comparison/summary.md). Lower is better.

adaptive − X	$\lvert\hat{\theta} - \theta^*\rvert$	Concept Brier	Held-out log loss
fixed order	-0.064 [-0.092, -0.038] **better**	-0.003 [-0.007, $+0.001$] *n.d.*	-0.009 [-0.016, -0.003] **better**
random	-0.082 [-0.110, -0.056] **better**	-0.006 [-0.010, -0.002] **better**	-0.016 [-0.024, -0.008] **better**

The CAT is more accurate in 5 of 6 comparisons; Brier score against fixed order shows no difference. The gains are small, and because the variance stop never fires with a 24-item bank (B5), the CAT is not shorter than the baselines.

5.3 Integrated Outcomes (Q3)

TABLE 12. Concepts mastered and retained recall, adaptive_full vs both baselines, 1,000 paired learners per cell.

Scen.	Metric	Static	Greedy	Adapt.	Adaptive − static [95% CI]	Adaptive − greedy [95% CI]
Toy S0	mastered (of 4)	2.036	2.036	2.092	+0.056 [0.040, 0.074]	+0.056 [0.040, 0.074]
	retained recall	0.636	0.636	0.660	+0.024 [0.017, 0.032]	+0.024 [0.017, 0.032]
Toy S1	mastered (of 4)	1.587	1.594	1.632	+0.045 [0.022, 0.064]	+0.038 [0.000, 0.075] *n.d.*
	retained recall	0.516	0.518	0.531	+0.015 [0.007, 0.024]	+0.013 [0.003, 0.023]
Toy S3	mastered (of 4)	2.334	2.334	2.451	+0.117 [0.096, 0.139]	+0.117 [0.096, 0.139]
Maths S0	mastered (of 16)	6.254	6.337	6.348	+0.094 [0.047, 0.140]	+0.011 [−0.031, 0.059] *n.d.*
	retained recall	0.589	0.603	0.684	+0.095 [0.087, 0.103]	+0.081 [0.074, 0.089]
Maths S3	mastered (of 16)	6.776	7.098	7.503	+0.727 [0.631, 0.824]	+0.405 [0.310, 0.500]
	retained recall	–	–	0.864	+0.104 [0.100, 0.109]	+0.081 [0.077, 0.085]

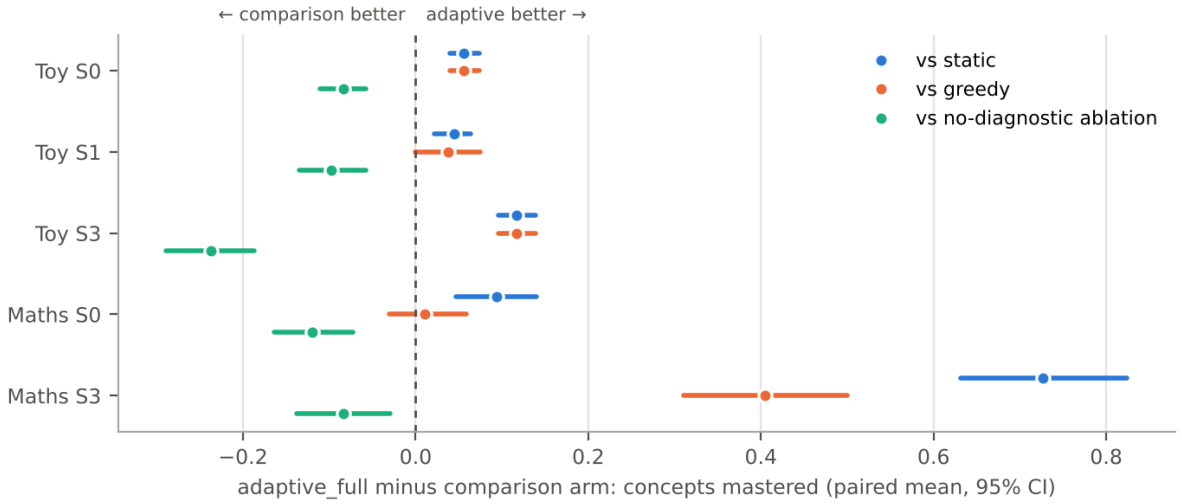


FIGURE 9. Paired differences in concepts mastered (adaptive_full minus each arm), 95% bootstrap CIs. Blue and orange: the two baselines. Aqua: the cold-start ablation, which now does better than the full system (Section 5.7).

Observations. The adaptive arm beats static in every scenario and on both outcomes. Against greedy it wins on mastery in 3 of 5 runs and on retained recall in all runs. On the matched simulator the effect is small (+0.056 of 4 concepts, about 3%). Under S1 forgetting costs every arm about 0.45 concepts and the adaptive advantage shrinks. The largest effects are under S3, which rewards practice directly.

Ablations. On toy S0: `no_replan` reaches 2.035 (adaptive better by 0.057); `no_difficulty_step`, `no_review` and `no_remediation` equal adaptive_full on mastery; `no_diagnostic` reaches 2.176, better than adaptive (−0.084 [−0.111, −0.058]). Under S1 only, the review ablation is slightly better (−0.017 [−0.033, −0.001]). So only the whole adaptive arm can be credited against the baselines, not its individual parts (B3).

5.4 Robustness (Q4)

TABLE 13. Adaptive − static, concepts mastered: curriculum × simulator (evaluation/results/w8/final_v2/before_after.txt).

	S0 matched	**S3 mismatched**
Toy, 4 concepts	+0.056 [0.040, 0.074]	+0.117 [0.096, 0.139]
Maths, 16 concepts	+0.094 [0.047, 0.140]	+0.727 [0.631, 0.824]

The advantage over static holds in all four cells. Because S3 rewards the adaptive arm's extra practice, "survives a mismatched simulator" is supported; "larger with real learners" is not.

5.5 Calibration of Mastery Estimates (Q5)

Run	**adaptive_full Brier [95% CI] / ECE**	**no-diagnostic**
Toy S0	0.136 [0.129, 0.143] / 0.082	0.146 / 0.064
Toy S1	0.196 [0.187, 0.203] / 0.163	0.189 / 0.145
Toy S3	0.167 [0.158, 0.176] / 0.089	0.142 / 0.079
Maths S0	0.190 [0.187, 0.193] / 0.166	0.187 / 0.141
Maths S3	0.188 [0.185, 0.192] / 0.155	0.191 / 0.156

TABLE 14. Mastery Brier score and ECE of the engine's p at the horizon against latent mastery.

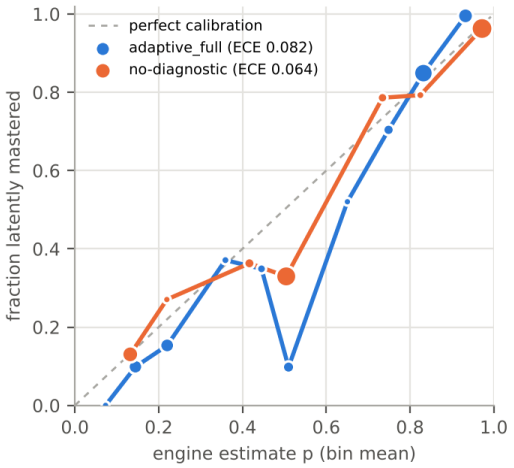


FIGURE 10. Reliability diagram, toy S0 (dot size = number of learner × concept pairs).

Static and greedy arms never update p, so calibration does not apply to them. High-confidence bins are well calibrated ($p \approx 0.93 \rightarrow 99.5\%$ mastered); the worst bin is $p \approx 0.5$, which holds concepts the learner never practised. Calibration degrades under forgetting and on the larger curriculum.

5.6 Hard Constraints (Q6)

0 prerequisite violations and 0 budget violations in all 9 arms of all 5 final runs (45,000 learner-runs), recounted independently of the planner; trace coverage is 100% for the four primary arms (the harness does not record coverage for ablations).

5.7 Failures and Edge Cases

- **Deadline-infeasible days.** 623 of 5,000 adaptive days on toy S0 and 5,751 of 10,000 on maths S0 cannot reach the target by the deadline. The planner returns the proof and a best-effort plan, so these are not idle days.
- **History: adaptive was once worse than static.** In Week 5 adaptive lost to static (−0.080); in Week 7, 4,000 of 5,000 adaptive days had no plan because completed-but-Uncertain prerequisites blocked their dependents. The deadlock fix moved adaptive − static from −0.102 to +0.140 (docs/w7/deadlock_fix_results.md); the later B10 fix reduced it to +0.056.
- **Edge cases covered by tests.** Cyclic, self-loop, missing-node and non-finite graphs are rejected; a block longer than the budget is proved infeasible; a missed day keeps completed work; a budget

cut gives a valid plan or a structured infeasibility trace; an Uncertain prerequisite never unlocks a dependent; alternating answers stay Developing; the planner cannot import latent state.

6. Milestone Verification Matrix

Full criteria are in docs/ACCEPTANCE_GROUPS.md. Recorded suite at the evidence revision: **299 passed** on Windows, Python 3.11 (weekly/W8.md). Rows without a group ID are requirements from the brief verified by experiment.

TABLE 15. Verification matrix.

Mile.	Test	Expected	Observed	Status	Evidence
M1	M1-a	Malformed graphs rejected (6 checks)	Rejected with error codes	Pass	tests/test_m1_curriculum_graph.py
M1	M1-b	BKT matches pyBKT 1.4.3 within 10^{-4}	Parity on all sequences	Pass	tests/test_m1_bkt_parity.py
M1	M1-c	Uncertain if $n < 3$ or $\sigma^2 > 0.05$; window gate	Boundary tests pass	Pass	tests/test_m1_learner_state.py
M1	M1-d	Zero prerequisite violations	0 in 45,000 learner-runs	Pass	tests/test_m3_states_feasibility.py
M2	M2-a	Max-information selection; stop at Var ≤ 0.15 or 10 items	Selection correct; variance stop never fires (B5)	Pass*	tests/test_m2_adaptive_diagnostic.py
M2	M2-b	$\hat{\theta} \rightarrow$ concept priors; unprobed = Beta(1,1)	Implemented incl. ADR-009 amendment	Pass	tests/test_w8_diagnostic_prior.py
M2	–	CAT compared with fixed and random	Better in 5 of 6 paired comparisons	Pass	evaluation/results/w8/diagnostic_comparison
M3	M3-a	Budget never exceeded, $B \in \{5, 10, 15, 30\}$	0 violations	Pass	tests/test_m3_path_planning.py
M3	M3-b	Greedy mean ratio ≥ 0.95 over 250	0.9737; 12/18 case reproduced	Pass	evaluation/results/w3/w3_summary.md
M3	M3-c	Oracle order-invariant, $N \leq 20$, timeout	Pass incl. brute-force check	Pass	tests/test_m3_oracle_order.py
M3/M6	M4-a	FEASIBLE / INFEASIBLE_ PROVED with repairs / HEURISTIC_NO_ PLAN	No repair synthesis for HEURISTIC_NO_ PLAN	Partial†	tests/test_m3_path_planning.py
M4	M4-b	2-up/1-down difficulty within [0.05, 0.95]	As specified	Pass	tests/test_m4_difficulty_adaptation.py
M5	M5-a	Two failed assessments trigger remediation	Pass in unit and scenario tests; never fires in the loop (B2)	Pass*	tests/test_m5_spaced_review_remediation.py

Mile.	Test	Expected	Observed	Status	Evidence
M5	M5-b	Review when $R < 0.85$, by risk $\times$ importance, within budget	Implemented and traced	Pass	tests/test_w8_review_importance.py
M6	M6-a	Events replan without losing completed work	As specified	Pass	tests/test_m6_replanning_events.py
M6	M6-b	Budget cut $\rightarrow$ valid plan or infeasibility with alternatives	As specified	Pass	tests/test_m6_dynamic_replanning.py
M7	M7-a	Trace coverage 1.0, 8-field schema	100% for primary arms	Pass	tests/test_m7_trace_contract.py
M7	M7-b	Blocked and skipped nodes carry reason codes	As specified	Pass	tests/test_m7_explainability.py
M8	M8-a	Different profiles $\rightarrow$ different plans	Novice and intermediate plans differ	Pass	tests/test_m8_evaluation.py
M8	M8-b	One command, exit code 0	`clean_run.ps1` PASS (W8)	Pass	evaluation/results/w8_evidence/clean_run_final.txt
M8	M8-c	Latent state invisible to planner	Import checks pass	Pass	tests/test_simulation.py
M8	M8-d	Retained recall vs baselines, matched and mismatched	Measured in S0, S1, S3	Pass	tests/test_beta_b3_retention.py
M8	–	Mastery prediction / calibration	Brier score and ECE reported	Pass	tests/test_w8_calibration.py

* Passes as written, with a behavioural caveat (Section 8). † Partial status agreed with the mentor in Week 4.

Approved changes. M4-a downgraded to partial; the universal 95% greedy guarantee replaced by an empirical distribution target (M3-b); the mismatched-simulator half of M8-d delivered with PFA (S3) in Week 8.

7. Mentor Feedback and Engineering Evolution

7.1 Feedback Closure

The repository records feedback by review round. W1-R items are the Week 1 review issues (weekly/W2.md, Section 2); FB006 and W2-FB004 are the IDs recorded in the Week 4 and Week 5 reports.

TABLE 16. Mentor feedback and responses.

FB-ID	Mentor request	Response	Change / justification	Evidence
W1-R1	Close the "Uncertain" loophole	Implemented	Uncertain if $n < 3$ or $\sigma^2 > 0.05$; 4-answer window gate	ADR-005, ADR-007

FB-ID	Mentor request	Response	Change / justification	Evidence
W1-R2	Uncertainty proxy is uncalibrated	Justified alternative	σ^2 declared as an evidence-count proxy; calibration of p now measured (Section 5.5)	ADR-005
W1-R3	Impossible profile ($p = 0.01$, $\sigma^2 = 0.25$)	Implemented	Valid parameters plus consistency check	config/profiles.yaml
W1-R4	Premature Weak classification	Implemented	Weak needs $p < 0.40$, evidence and 3 of 4 incorrect	ADR-007
W1-R5	Unsupported universal 95% greedy claim	Implemented	Claim withdrawn; 250-instance distribution and 12/18 counterexample	docs/METRICS.md
W1-R6	Truncated test-case register	Implemented	Full TC-1.1 … TC-8.2 traceability	TRACEABILITY.md
W1-R7	Module boundaries conflated	Implemented	Eight modules M1–M8 in code and tests	README.md
W1-R8	Retention on held-out outcomes	Partial	Real-data route deferred; delayed quiz on simulated learners	docs/DATA_STRATEGY.md
W1-R9	Separate infeasibility reasons	Implemented	Block, capacity and heuristic outcomes distinguished	src/planning/feasibility.py
W1-R10	Provenance and checksums	Implemented	SHA-256 config hash and run manifests	src/provenance.py
W2-FB004	(Week 2 feedback item)	Not traceable	Original evidence recorded as “not found” in the W4/W5 reports	weekly/W4.md
FB006	Clean-clone reproducibility	Implemented	`clean_run.ps1` passes on a fresh clone (W4: 202 passed; W8: PASS)	weekly/W4.md
Mentor (W4)	Accept partial M4-a, M8-a, M8-d	Two of three closed	M8-a passing since W6; M8-d since W8 (S3); M4-a still partial	docs/ACCEPTANCE_GROUPS.md

7.2 Defects Found and Fixed

TABLE 17. Main defects; each was reproduced by a failing test before the fix (before/after evidence under evaluation/results).

When	Defect	Fix
W3 (10)	Planned prerequisite treated as completed or mastered; trace checker accepted invalid entries; oracle depended on input order; capacity check overstated feasibility; window size ignored; invalid BKT config fell back silently; non-finite values and cycles; remaining budget recorded after selection; unused reason codes; duplicate configs	Separate planned/completed/Mastered records; strict checker; order-independent oracle; witness-plan rule; config validation; sequential budget recording; provenance
W4 (8)	Assessment applied twice in the CLI; PolicyConfig dropped; silent policy loading; permissive trace types; edge NaN accepted; infeasible runs checked as feasible; truncated hash; profiles not expanded to concepts	Single application; policy propagated and schema-checked; checker bounds; edge validation; status-routed traces; full-precision hash; group mapping
W6	Multi-seed harness ran seed 42 only	All five seeds run and pooled
W7	Adaptive deadlock: completed-but-unproven prerequisites blocked dependents (4,000 of 5,000 no-plan days)	Practice-for-evidence; mid-horizon trace fix
W8	S1 never applied forgetting during study; planner idled on infeasible days; diagnostic gave $n = 0$ (B10); importance unused (B11)	Daily forgetting; best-effort plan; $n = k$ priors; importance-weighted review

7.3 Timeline

TABLE 18. Week-by-week increments (weekly).

Week	Main increment
W1	Problem boundary, assumptions, data strategy, literature comparison, formal models, metrics, traceability
W2	Graph loader, learner state, static planner, greedy and oracle, 250-instance benchmark
W3	Ten correctness repairs; pyBKT parity; replanning; methodology frozen; Mid-Review
W4	CAT diagnostic, difficulty, review and remediation; horizontal slice of all modules; eight repairs
W5	Integrated M1–M8 evaluation against both baselines (adaptive initially worse)
W6	Clean-run audit, multi-seed fix, S1/S2 configs, retention quiz, ownership packs
W7	Best-singleton fallback; website and live demos; adaptive deadlock fix
W8	S1 forgetting, best-effort plans, maths curriculum, mismatched simulator, final fixes, report

7.4 Methodology Record

The approach was **changed**. The change is recorded in decisions/METHODOLOGY_RECORD.md on **12 October 2026**, after the 9 October deadline; it is not backdated. **Old approach:** fit BKT, IRT and HLR on Junyi and ASSISTments logs. **New evidence:** the public logs have no latent ground truth, no prerequisite DAGs and no lesson durations, so they cannot evaluate a planner. **Alternatives:** real data only; a hybrid; simulator first. **Selected:** simulator first, with public data as optional secondary vali-

dation. **Impact:** outcome claims are limited to synthetic learners; retention is measured by a simulated delayed quiz; real-data validation moved to the way forward (Section 9).

8. Current State, Limitations and Risks

TABLE 19. What works, what is partial and what is missing.

Working	**Partial**	**Missing**
Graph validation; BKT state and classification; CAT diagnostic; static, greedy, best-singleton and oracle planners; feasibility proofs; replanning with best-effort plans; difficulty staircase; importance-weighted review; complete audit traces; S0/S1/S3 simulators; integrated evaluation with paired CIs; calibration; CLI; website with live demos.	Repair suggestions for HEURISTIC_NO_PLAN (M4-a); remediation works in tests but never triggers in the loop; the CAT variance stop never fires; the diagnostic is connected to planning but currently lowers mastery; S2 configured but not re-run.	Any real or public-data validation; fitted HLR half-lives; the bounded bandit policy; per-concept diagnostic estimates; a learner-facing interface.

TABLE 20. Open defects and risks (docs/final/LIMITATIONS.md).

ID	**Defect / risk**	**Effect**	**Workaround**
B12	Diagnostic lowers mastery: cold-start ablation beats adaptive_full in all five runs	The diagnostic cannot be credited	Reported; per-concept priors in Section 9
B1/B2	Remediation ablation disables nothing; remediation never fires in the loop	No integrated evidence for remediation	Unit and scenario tests; Section 9
B3	Difficulty, review and remediation ablations equal or nearly equal adaptive_full	No per-component attribution	Whole-arm claims only
B4	Deadline-infeasible days (toy 623 / 5,000; maths 5,751 / 10,000)	Target unreachable for many learners	Proof plus best-effort plan
B5	CAT stop rule never fires with 24 items	Diagnostic length fixed at 10	Larger bank (Section 9)
B6/B7	Static baseline can miss a feasible plan (3 / 3,000) and has no best-effort plan	Baseline slightly weaker by design	Disclosed
B8	Greedy not optimal (worst ratio 0.214)	Some plans below optimum	Best-singleton fallback
B9	Planner does not model forgetting directly	Lost mastery recovered only through review	Measured in S1
B13	Baselines keep $n = 0$ diagnostic priors	Adaptive receives evidence the baselines do not	Disclosed; baselines frozen

ID	Defect / risk	Effect	Workaround
Validity	Synthetic learners; S0 matches the planner; S3 favours practice; hand-made curricula	External validity unknown	Mismatched run; real data in Section 9

9. Way Forward

TABLE 21. Prioritised next tasks.

Prio.	Task	Starting point	Risk / dependency	Done when
P1	Per-concept diagnostic priors so the diagnostic helps instead of hurting (B12)	`init_from_diagnostic` in src/diagnostic/diagnostic.py; ADR-009	Needs several items per concept; may change headline results	adaptive_full − ablation_no_diagnostic CI ≥ 0 on toy S0 and S1; hard constraints unchanged
P2	Make remediation reachable and the remediation ablation real (B1, B2)	`get_remediation_ancestors` in src/planning/review.py; arm handling in `integrated.py`	Changes adaptive behaviour; re-run all scenarios	Remediation fires in toy S1; the ablation differs from adaptive_full
P3	Validate on a public dataset: fit BKT and HLR, check calibration	docs/DATA_STRATEGY.md, examples/pybkt_fit_predict.py	Licence and schema; no DAG in raw logs	Held-out log loss and ECE for fitted vs default parameters
P4	Re-run the difficulty-sensitive scenario S2	experiments/beta_s2.yaml	Low	Pooled S2 results with paired CIs; difficulty ablation compared
P5	Repairs for HEURISTIC_NO_PLAN (close M4-a); larger item bank so the CAT stop works (B5)	src/planning/feasibility.py, scripts/make_item_bank.py	Search cost on large DAGs	M4-a passes as written; mean CAT length < 10 at equal accuracy

10. Contributions, AI Use and Attribution

TABLE 22. Contributions (git history at the evidence revision).

Member	Areas and modules	Representative commits
Bhuvan Chandra (Track A)	Curriculum graph and validation; static, greedy and oracle planners; feasibility; replanning and best-effort plans; trace schema and checker; CLI; S1 forgetting fix; maths curriculum and S3 harness; final fixes B10/B11, diagnostic comparison and calibration; final report; website curriculum, S1 and robustness pages. 101 non-merge commits.	`0549b2b`, `0262d45`, `01fc631`, `840a12b`, `72b8f02`, `9b50338`, `70183b7`, `edde468`, `b2994f6`, `2357507`

Member	Areas and modules	Representative commits
Abdur Rehman (Track B)	BKT parity tests against pyBKT; CAT diagnostic and synthetic 2PL item bank; comparative evaluation; simulator variants and retention quiz; multi-seed harness; best-singleton fallback; adaptive deadlock fix; project website and Pyodide demos. 73 non-merge commits.	`34a4946`, `339711c`, `a20513a`, `c6f59e5`, `572850f`, `d7e1ea2`, `c5d1a8f`, `f86a1c1`

AI-assisted work. Coding agents were used during development to support implementation, debugging, and refinement across different project phases. **How it was checked:** Changes were reviewed against project requirements, validated through testing, and evaluated using recorded experimental results. The final implementation was verified through the test suite and evaluation metrics, with reported results reflecting the observed outcomes.

Third-party assets. Only the open-source libraries in Section 2.3, used unmodified under BSD-3, MIT and MPL-2.0 licences. All curricula, item banks and learners are synthetic and generated by the team, with provenance files beside each data file. No private learner data or unlicensed assets are included.

References

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[10] Software: NetworkX 3.1, pyBKT 1.4.3, catsim 0.21.0, NumPy, pandas, SciPy, scikit-learn, pytest, Pyodide 0.26.4.

A. Sample Plan and Trace

From the horizontal slice, learner 1, day 1, toy curriculum, 10 min/day, target D (evaluation/results/w4/slice/plan_learner1_day1.json). B, C and D are *conditional*: they follow a prerequisite planned on an earlier day and are re-checked before they are done; C is also an assessment candidate.


```
{ "status": "FEASIBLE",
  "feasibility_explanation": { "status": "FEASIBLE", "total_capacity": 30, "w_req": 30 },
  "assessment_candidates": ["C"],
  "days": [
    { "day": 1, "activities": ["A", "B"], "conditional_activities": ["B"],
      "minutes_used": 10, "minutes_remaining": 0 },
    { "day": 2, "activities": ["C"], "conditional_activities": ["C"],
      "minutes_used": 10, "minutes_remaining": 0 },
    { "day": 3, "activities": ["D"], "conditional_activities": ["D"],
      "minutes_used": 10, "minutes_remaining": 0 } ] }
```


One trace entry (the first of 12 for this plan: 4 candidates × 3 days):


```
{ "day": 1, "node_id": "A", "decision": "SELECTED", "reason_code": "SELECTED",
  "concept_state": { "class": "Uncertain", "p": 0.5465, "sigma2": 0.0826,
                     "n": 0, "kappa": 2 },
  "constraints": { "budget": 10, "remaining_budget": 10, "prerequisites": {},
                   "conditional": false, "revalidate_before_execution": false },
  "score_terms": {}, "policy_version": "static_baseline_v0" }
```


B. Interface Screenshots

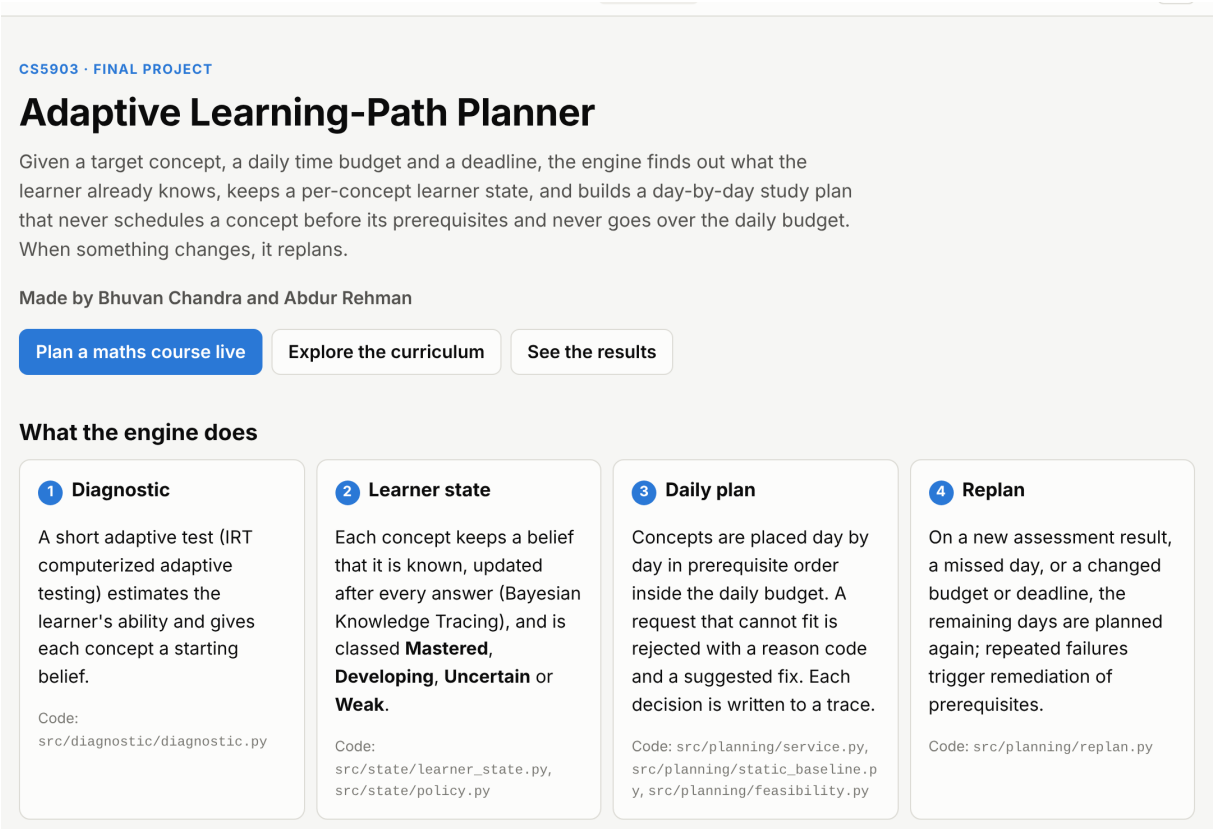


FIGURE 11. Overview page: what the engine does, with the code behind each step.

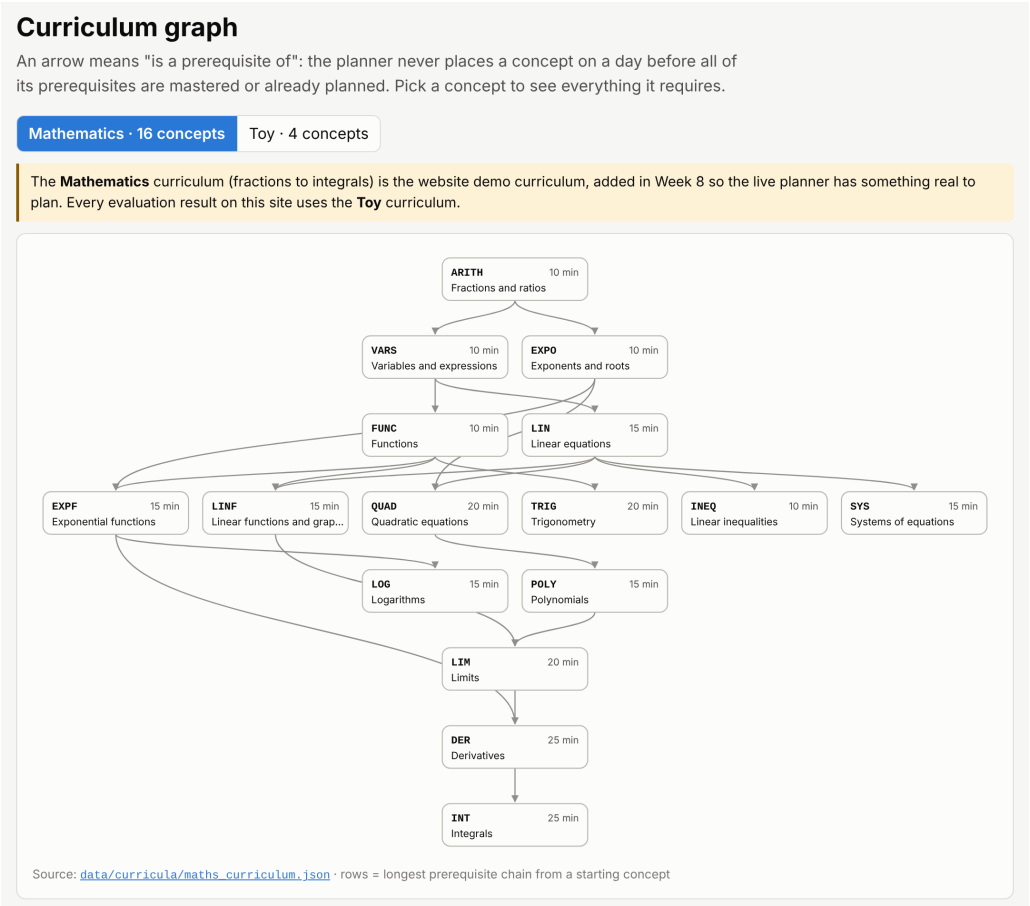


FIGURE 12. Curriculum page: the 16-concept maths DAG with durations; clicking a concept highlights its prerequisites.

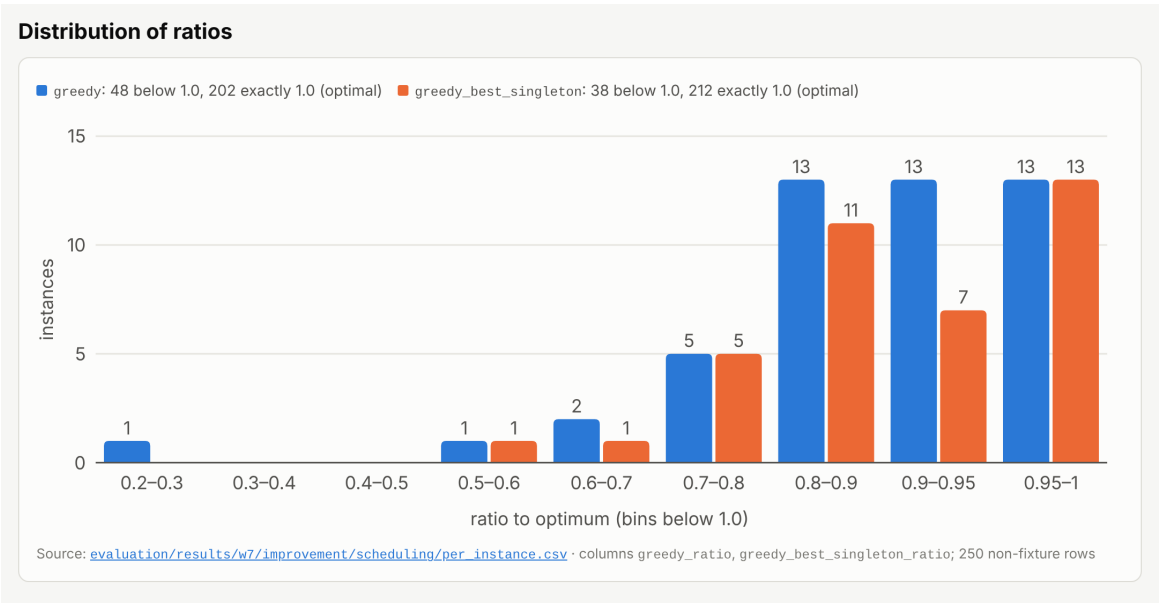


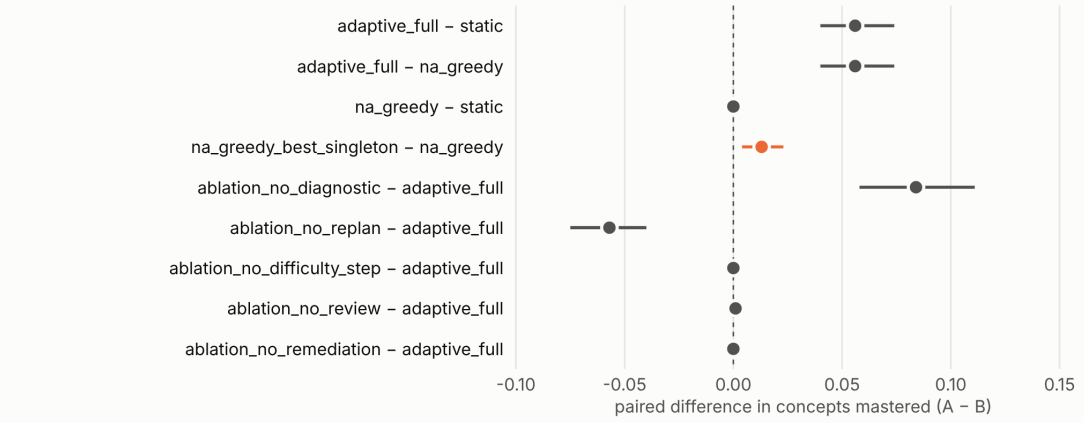
FIGURE 13. Scheduling benchmark page: distribution of greedy and best-singleton ratios over 250 instances.

Paired differences in concepts mastered

Each learner is run under both arms, so differences are taken learner by learner and bootstrapped. An arm is called better only if the 95% CI excludes zero.

Comparison (A − B)	Mean diff.	95% CI	Learners changed (+ / −)	Claim
adaptive_full − static_prerequisite_order	+0.0560	[0.0400, 0.0740]	73 (+63 / −10)	adaptive_full better (higher mastery)
adaptive_full − non_adaptive_greedy	+0.0560	[0.0400, 0.0740]	73 (+63 / −10)	adaptive_full better (higher mastery)
non_adaptive_greedy − static_prerequisite_order	0.0000	[0.0000, 0.0000]	0 (+0 / −0)	no difference shown
non_adaptive_greedy_best_singleton − non_adaptive_greedy	+0.0130	[0.0040, 0.0230]	23 (+18 / −5)	non_adaptive_greedy_best_singleton better (higher mastery)
ablation_no_diagnostic − adaptive_full	+0.0840	[0.0580, 0.1110]	169 (+123 / −46)	ablation_no_diagnostic better (higher mastery)
ablation_no_replan − adaptive_full	-0.0570	[-0.0750, -0.0400]	72 (+9 / −63)	adaptive_full better (higher mastery)
ablation_no_difficulty_step − adaptive_full	0.0000	[0.0000, 0.0000]	0 (+0 / −0)	no difference shown
ablation_no_review − adaptive_full	+0.0010	[0.0000, 0.0030]	1 (+1 / −0)	no difference shown
ablation_no_remediation − adaptive_full	0.0000	[0.0000, 0.0000]	0 (+0 / −0)	no difference shown

Source: evaluation/results/w8/final_v2/integrated_s0/per_learner_seed_*.csv · comparisons are the ones each summary_seed_<s>.md reports (sections 3 and 4), pooled over 1,000 learners



Labels shorten non_adaptive_ to na_ and static_prerequisite_order to static.

Source: evaluation/results/w8/final_v2/integrated_s0/per_learner_seed_*.csv

FIGURE 14. Results page: paired differences in concepts mastered for S0, with the claim for each comparison.

C. Reproduction and Evidence Links

All at revision d974503. Final report and limitations: docs/final/FINAL_REPORT.md, docs/final/LIMITATIONS.md. Demo script: docs/final/DEMO.md. Formal models and metrics: docs/FORMAL_MODELS.md, docs/METRICS.md. Acceptance groups: docs/ACCEPTANCE_GROUPS.md. Decisions: decisions. Weekly records: weekly. Raw results: evaluation/results; final runs evaluation/results/w8/final_v2; diagnostic comparison evaluation/results/w8/diagnostic_comparison. Before and after evidence: evaluation/results/w8_evidence. Website source: site; live site bhuvan-56.github.io/adaptive-learning-path-planner-DAT/site/.

No optional bonus (BP-ID) claims are made.